Python: exercises on iteration

This notebook was generated in **solutions** mode.

Exercise: Print Numbers from 1 to N

Exercise Description

Write a function that prints all numbers from 1 to N using a for loop and range.

Your solution

```
def print_numbers(n):
    """Print all numbers from 1 to n."""
    for i in range(1, n + 1):
        print(i)
```

Test your solution

```
# Test case 1: Test with small number
print_numbers(5)
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Sum of First N Natural Numbers

Exercise Description

Write a function to calculate the sum of the first N natural numbers (1 + 2 + 3 + ... + N) using a while loop.

Your solution

```
def sum_natural_numbers(n):
    """Calculate sum of first n natural numbers."""
    i = 1
    total = 0
    while i <= n:
```

```python
        total += i
        i += 1
    return total
```

Test your solution

```python
# Test case 2: Test with another number
result = sum_natural_numbers(3)
expected = 1 + 2 + 3  # 6
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_natural_numbers(3) = {result}")
```

Test your solution

```python
# Test case: Example from solution
result = sum_natural_numbers(5)
print(f"Sum of first 5 natural numbers: {result}")  # 15
```

Debug your solution

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```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Print Even Numbers from 2 to 20

Exercise Description

Write a function that prints all even numbers between 2 and 20 using a for loop and range.

Your solution

```
def print_even_numbers():
    """Print even numbers from 2 to 20."""
    for i in range(2, 21, 2):
        print(i)
```

Test your solution

```
# Test case 2: Verify range and step
# Expected: prints 2, 4, 6, 8, 10, 12, 14, 16, 18, 20 (each on new line)
print_even_numbers()
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Count Down from N to 1 (If Even)

Exercise Description

Write a function to print numbers from N to 1 using a while loop, but only if N is even.

Your solution

```
def countdown_if_even(n):
    """Count down from n to 1, but only if n is even."""
    if n % 2 == 0:
        while n > 0:
            print(n)
            n -= 1
```

Test your solution

```
# Test case 2: Test with odd number (should print nothing)
countdown_if_even(3)
```

Test your solution

```
# Test case 2 (assert)
countdown_if_even(6)  # Prints 6, 5, 4, 3, 2, 1
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Multiplication Table of 3

Exercise Description

Write a function that prints the multiplication table of 3 (from 3x1 to 3x10) using a for loop.

Your solution

```
def multiplication_table_3():
    """Print multiplication table of 3."""
    for i in range(1, 11):
        print(f"3 x {i} = {3 * i}")
```

Test your solution

```
# Test case 2: Verify format
# Expected: prints 3 x 1 = 3, 3 x 2 = 6, ..., 3 x 10 = 30
```

```
multiplication_table_3()
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Sum of First N Even Numbers

Exercise Description

Write a function to calculate the sum of the first N even numbers (2 + 4 + 6 + ...) using a while loop.

Your solution

```
def sum_first_n_even(n):
    """Calculate sum of first n even numbers."""
    i = 2
    total = 0
    count = 0
    while count < n:
        total += i
```

```
        i += 2
        count += 1
    return total
```

Test your solution

```
# Test case 2: Test with another number
result = sum_first_n_even(5)
expected = 2 + 4 + 6 + 8 + 10  # 30
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_first_n_even(5) = {result}")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Numbers Divisible by 5 (1 to 50)

Exercise Description

Write a function that prints all numbers divisible by 5 between 1 and 50 using a for loop and range.

Your solution

```
def print_divisible_by_5():
    """Print numbers divisible by 5 between 1 and 50."""
    for i in range(5, 51, 5):
        print(i)
```

Test your solution

```
# Test case 2: Verify range
# Expected: prints 5, 10, 15, 20, 25, 30, 35, 40, 45, 50
print_divisible_by_5()
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Factorial of 5

Exercise Description

Write a function to find the factorial of 5 using a while loop.

Your solution

```
def factorial_5():
    """Calculate factorial of 5."""
    i = 5
    factorial = 1
    while i > 0:
        factorial *= i
        i -= 1
    return factorial
```

Test your solution

```
# Test case 2: Verify result
result = factorial_5()
assert result == 120, f"Expected 120, got {result}"
print(f"✓ Factorial of 5 is {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Sum of Odd Numbers (1 to 20)

Exercise Description

Write a function to calculate the sum of all odd numbers between 1 and 20 using a for loop.

Your solution

```
def sum_odd_numbers():
    """Calculate sum of odd numbers between 1 and 20."""
    total = 0
    for i in range(1, 21, 2):
        total += i
    return total
```

Test your solution

```
# Test case 2: Verify result
result = sum_odd_numbers()
assert result == 100, f"Expected 100, got {result}"
print(f"✓ Sum of odd numbers 1-20 is {result}")
```

Test your solution

```
# Test case: Example from solution
result = sum_odd_numbers()
```

```
print(f"Sum of odd numbers 1-20: {result}")  # 100
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Numbers Divisible by K (1 to N)

Exercise Description

Write a function to print numbers between 1 and N that are divisible by K using a for loop and range.

Your solution

```
def print_divisible_by_k(n, k):
    """Print numbers between 1 and n that are divisible by k."""
    for i in range(k, n + 1, k):
        print(i)
```

Test your solution

```
# Test case 2: Test with n=20, k=4
# Expected: prints 4, 8, 12, 16, 20
print_divisible_by_k(20, 4)
```

Test your solution

```
# Test case: Example from solution
print_divisible_by_k(20, 3)  # Prints 3, 6, 9, 12, 15, 18
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Count Digits in a Number

Exercise Description

Write a function that counts how many digits are in a positive integer.

Your solution

```python
def count_digits(num):
    """Return the number of digits in positive integer num."""
    if num == 0:
        return 1
    count = 0
    while num > 0:
        num //= 10
        count += 1
    return count
```

Test your solution

```python
assert count_digits(123456) == 6
assert count_digits(7) == 1
assert count_digits(10) == 2
print("✓ count_digits tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Cubes from 1 to 5

Exercise Description

Write a function that returns a list with the cubes of numbers from 1 to 5 using a for loop.

Your solution

```
def cubes_1_to_5():
    result = []
    for i in range(1, 6):
        result.append(i**3)
    return result
```

Test your solution

```
assert cubes_1_to_5() == [1, 8, 27, 64, 125]
print("✓ cubes_1_to_5 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Product of Numbers from 1 to N

Exercise Description

Write a function that returns the product of numbers from 1 to N using a for loop.

Your solution

```
def product_1_to_n(n):
    product = 1
    for i in range(1, n + 1):
        product *= i
    return product
```

Test your solution

```
assert product_1_to_n(1) == 1
assert product_1_to_n(3) == 6
assert product_1_to_n(5) == 120
print("✓ product_1_to_n tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Odd Numbers from 1 to 15

Exercise Description

Write a function that returns all odd numbers from 1 to 15 using a while loop.

Your solution

```
def odd_1_to_15():
    result = []
    i = 1
    while i <= 15:
        result.append(i)
        i += 2
    return result
```

Test your solution

```
assert odd_1_to_15() == [1, 3, 5, 7, 9, 11, 13, 15]
print("✓ odd_1_to_15 tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Print Numbers from 5 to 1

Exercise Description

Write a function that returns numbers from 5 to 1 using a for loop with range.

Your solution

```
def countdown_5_to_1():
    return [i for i in range(5, 0, -1)]
```

Test your solution

```
assert countdown_5_to_1() == [5, 4, 3, 2, 1]
print("✓ countdown_5_to_1 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Sum of the First 7 Odd Numbers

Exercise Description

Write a function to find the sum of the first 7 odd numbers using a while loop.

Your solution

```
def sum_first_7_odds():
    i = 1
    count = 0
    total = 0
    while count < 7:
        total += i
        i += 2
        count += 1
    return total
```

Test your solution

```
assert sum_first_7_odds() == 49
print("✓ sum_first_7_odds tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Repeat Until ”stop”

Exercise Description

Write a function that, given a sequence of strings, counts how many entries occur until the string "stop" appears.

Your solution

```
def until_stop_count(inputs):
    """Return the number of entries until "stop" appears (case-sensitive)."""
    count = 0
    for s in inputs:
        if s == "stop":
```

```
            break
        count += 1
    return count
```

Test your solution

```
assert until_stop_count(["a", "b", "stop", "x"]) == 2
assert until_stop_count(["stop"]) == 0
assert until_stop_count(["hello", "world"]) == 2
print("✓ until_stop_count tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Repeat String as Many Times as Its Length

Exercise Description

Write a function that takes a string and returns a list with the string repeated as many times as its length.

Your solution

```
def repeat_string_len_times(s):
    return [s for _ in range(len(s))]
```

Test your solution

```
assert repeat_string_len_times("abc") == ["abc", "abc", "abc"]
assert repeat_string_len_times("") == []
print("✓ repeat_string_len_times tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Ask for a Password

Exercise Description

Write a function that, given a sequence of password attempts, returns the number of attempts needed to enter the correct password ("password123"). Return -1 if never entered.

Your solution

```
def password_attempts(attempts, correct="password123"):
    count = 0
    for a in attempts:
        count += 1
        if a == correct:
            return count
    return -1
```

Test your solution

```
assert password_attempts(["x", "y", "password123"]) == 3
assert password_attempts(["password123"]) == 1
assert password_attempts(["a", "b"]) == -1
print("✓ password_attempts tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```